

Title: Git Hands-On Lab 4 - Rebasing and Cherry-Picking**Problem Statement :****Objectives:**

- Explain rebasing and cherry-picking
- Demonstrate resolving merge conflicts during rebase
- Use cherry-pick to apply specific commits

In this hands-on lab, you I learnt how to:

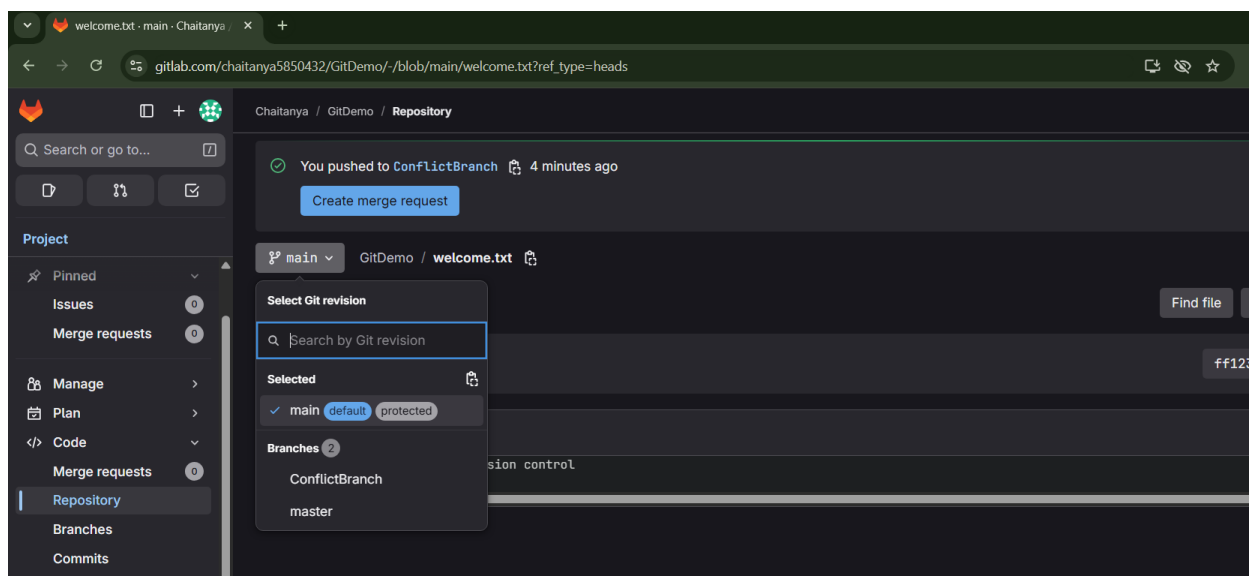
- Perform a rebase to maintain a linear history
- Resolve conflicts during rebase
- Use cherry-pick to apply selected commits to another branch

Prerequisites:

- Setting up Git environment with P4Merge tool for Windows
- Completion of Git Hands-On Lab 3

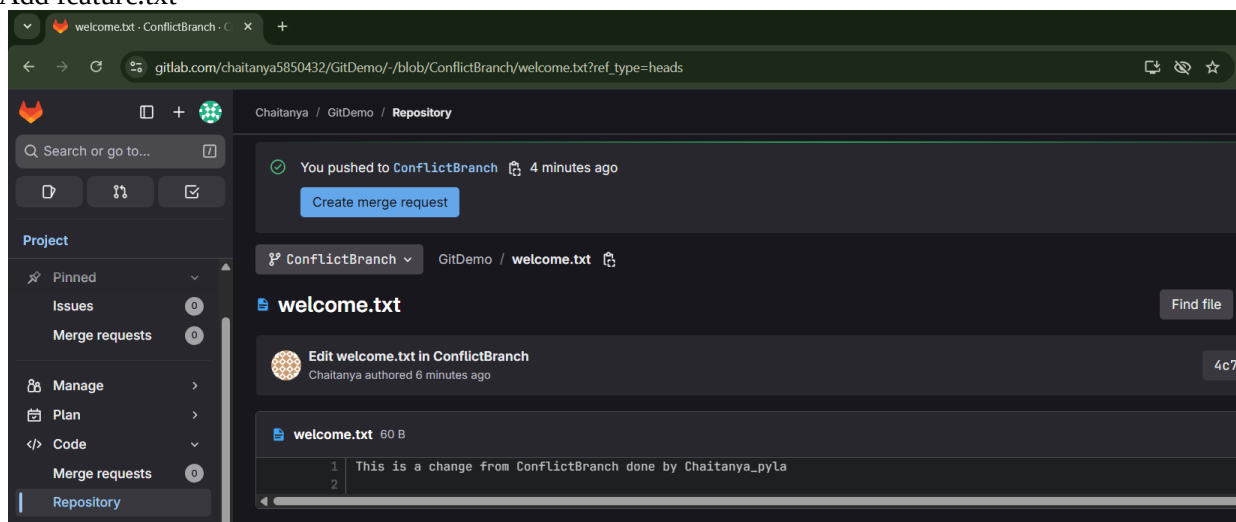
Rebasing:

1. Create a new branch “FeatureRebase” from the master branch.
Command: git branch FeatureRebase



2. Switch to the “FeatureRebase” branch and make a commit with a new file “feature.txt” containing “Feature content”.

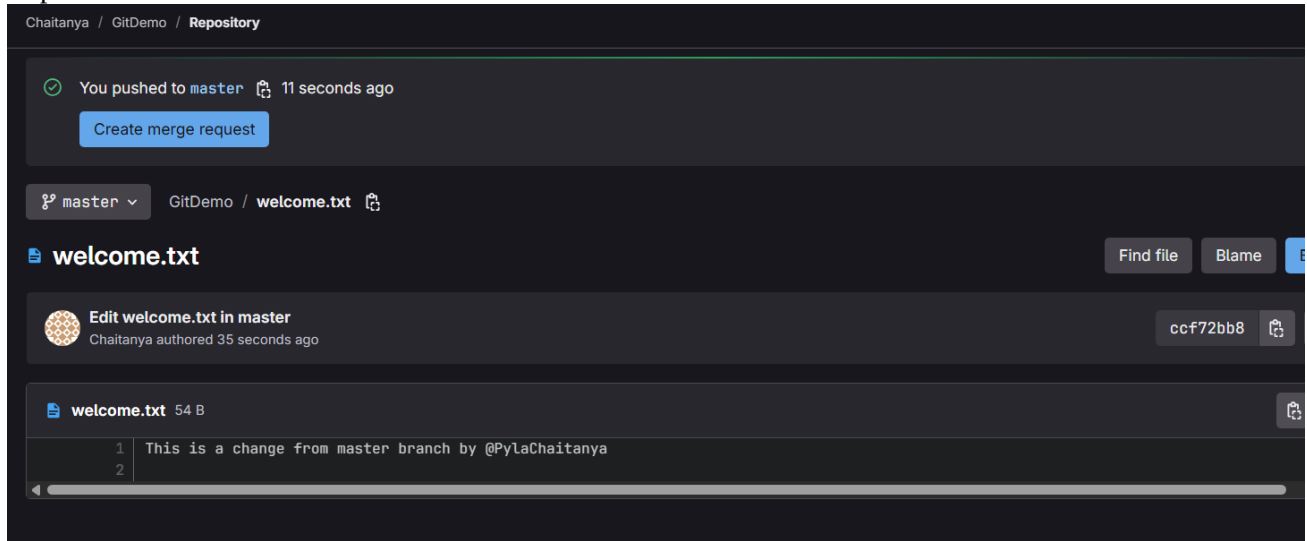
Commands: git checkout FeatureRebase; echo "Feature content" > feature.txt; git add feature.txt; git commit -m "Add feature.txt"



4. Git-HOL

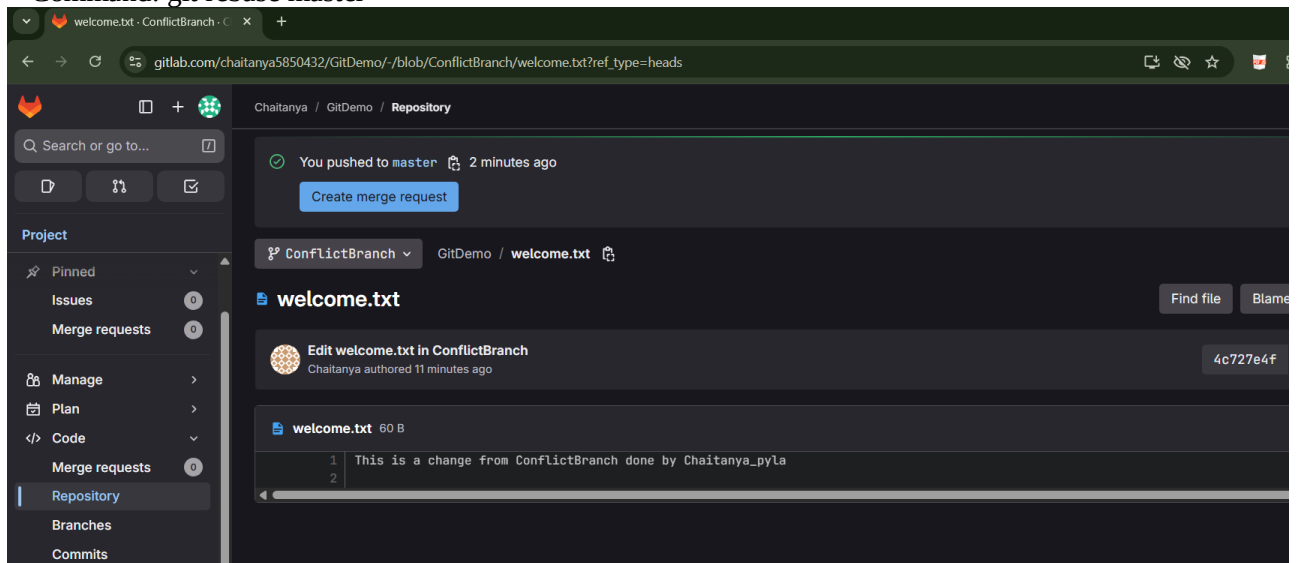
3. Switch back to the master branch and make a commit with changes to “branchfile.txt” (e.g., add “Master update”).

Commands: `git checkout master`; `echo "Master update" >> branchfile.txt`; `git add branchfile.txt`; `git commit -m "Update branchfile.txt"`



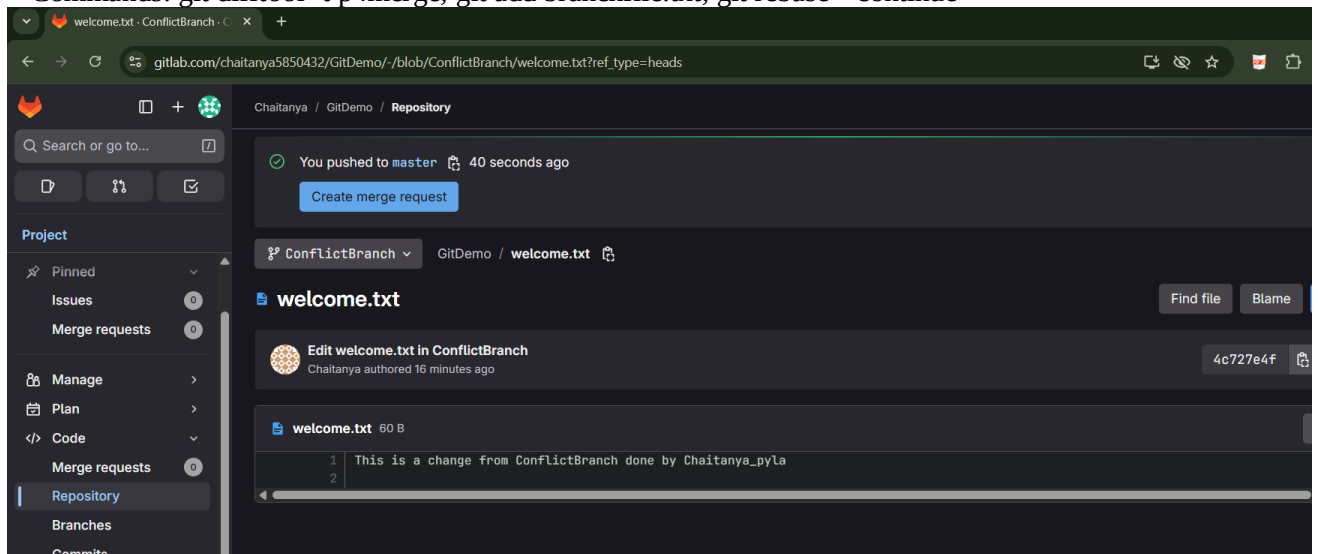
4. Switch to “FeatureRebase” and rebase it onto master.

Command: `git rebase master`



5. If a conflict occurs, resolve it using P4Merge, then continue the rebase.

Commands: `git difftool -t p4merge`; `git add branchfile.txt`; `git rebase --continue`



6. Verify the rebase with “git log --oneline --graph --decorate”.

Command: git log --oneline --graph --decorate

```
MINGW32/c/Users/pchai/GitDemo

pchai@Chaitanya MINGW32 ~/GitDemo (ConflictBranch)
$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 2 commits.
(use "git push" to publish your local commits)

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ echo "This is a change from master branch by @PyLaChaitanya">welcome.txt

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git add welcome.txt
warning: in the working copy of 'welcome.txt', LF will be replaced by CRLF th
e next time Git touches it

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git commit -m "Edit welcome.txt in master"
[master ccf72bb] Edit welcome.txt in master
1 file changed, 1 insertion(+), 1 deletion(-)

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 316 bytes | 316.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote:   https://gitlab.com/chaitanya5850432/GitDemo/-/merge_requests/new?me
rge_request%5Bsource_branch%5D=master
remote:
To https://gitlab.com/chaitanya5850432/GitDemo.git
22be986..ccf72bb master -> master

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git merge ConflictBranch
merge: ConflictBranch - not something we can merge

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ ^C
```

Cherry-Picking:

1. Create a new branch “CherryPickBranch” from master.

Command: git branch CherryPickBranch

```
MINGW32/c/Users/pchai/GitDemo

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git merge ConflictBranch
Auto-merging welcome.txt
CONFLICT (content): Merge conflict in welcome.txt
Automatic merge failed; fix conflicts and then commit the result.

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ git checkout master
error: you need to resolve your current index first
welcome.txt: needs merge

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ ^C

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ cat welcome.txt
<<<<<< HEAD
This is a change from master branch by @PyLaChaitanya
=====
This is a change from ConflictBranch done by Chaitanya_pyLa
>>>>>> ConflictBranch

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ notepad++ welcome.txt

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ cat welcome.txt
This is a change from master branch by @PyLaChaitanya

This is a change from ConflictBranch done by Chaitanya_pyLa

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ git add welcome.txt

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ git commit -m "Resolved merge conflict in welcome.txt"
[master 333c3f8] Resolved merge conflict in welcome.txt

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git push origin master
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
```

2. Switch to “CherryPickBranch” and cherry-pick the commit from “FeatureRebase” that added “feature.txt”.

Commands: git checkout CherryPickBranch; git cherry-pick <commit-hash>

```
MINGW32 ~/GitDemo
pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ cat welcome.txt
<<<<<< HEAD
This is a change from master branch by @PylaChaitanya
=====
This is a change from ConflictBranch done by Chaitanya_pyla
>>>>>> ConflictBranch

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ notepad++ welcome.txt

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ cat welcome.txt
This is a change from master branch by @PylaChaitanya

This is a change from ConflictBranch done by Chaitanya_pyla

pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ git add welcome.txt

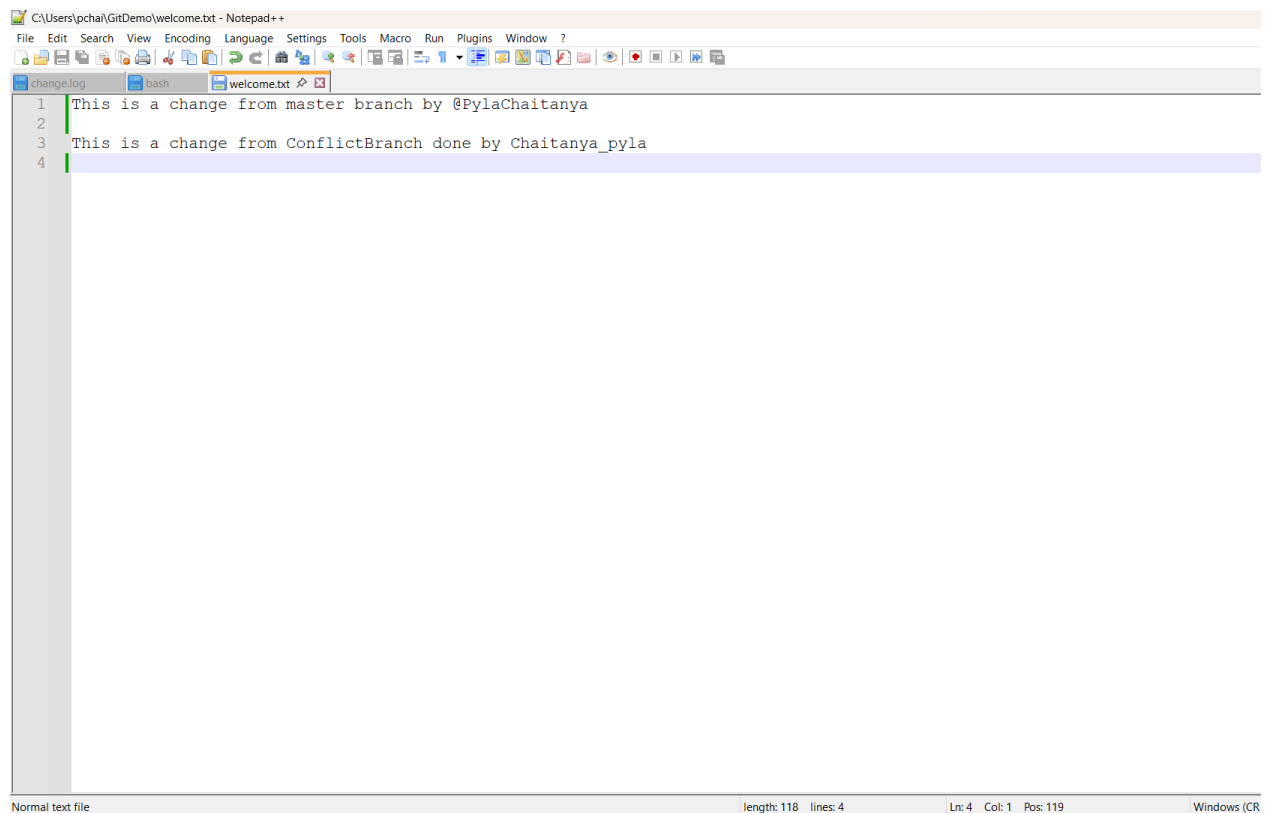
pchai@Chaitanya MINGW32 ~/GitDemo (master|MERGING)
$ git commit -m"Resolved merge conflict in welcome.txt"
[master 333c3f8] Resolved merge conflict in welcome.txt

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$ git push origin master
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 381 bytes | 381.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: To create a merge request for master, visit:
remote:   https://gitlab.com/chaitanya5850432/GitDemo/-/merge_requests/new?me
rge_request%5Bsource_branch%5D=master
remote:
To https://gitlab.com/chaitanya5850432/GitDemo.git
ccf72bb..333c3f8 master -> master

pchai@Chaitanya MINGW32 ~/GitDemo (master)
$
```

3. Verify the cherry-picked commit with “git log --oneline --graph --decorate”.

Command: git log --oneline --graph --decorate



```
C:\Users\pchai\GitDemo\welcome.txt - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
change.log bash welcome.txt
1 This is a change from master branch by @PylaChaitanya
2
3 This is a change from ConflictBranch done by Chaitanya_pyla
4

Normal text file
length: 118 lines: 4
Ln: 4 Col: 1 Pos: 119
Windows (CR)
```